Example 3.13

In an examination, a student scores 4 marks for every correct answer and loses one mark for every wrong answer. If he answers 60 questions in all and gets 130 marks, find the number of questions he answered correctly.

Solution Let the number of correct answers be x

Thus the number of wrong answers $= 60 - x$

$$\begin{aligned} \text{Then, } 4x - 1(60 - x) &= 130 \\ 4x - 60 + x &= 130 \\ 4x + x - 60 + 60 &= 130 + 60 \quad \text{[Add 60 on both sides]} \\ 5x + 0 &= 190 \\ 5x &= 190 \end{aligned}$$

Divide by 5 on both the sides,

$$\frac{5x}{5} = \frac{190}{5}$$
$$x = 38$$

Hence, the number of correct answer is 38.

Example 3.14

A school bus starts with full strength of 60 students. It drops some students at the first bus stop. At the second bus stop, twice the number of students get down from the bus. 6 students get down at the third bus stop and the number of students remaining in the bus is only 3. How many students got down at the first stop?

Solution

Since we do not know the number of students who get down at the first stop, let us take the number as x. The number of students get down at the second bus stop is $2x$.



Fig.3.9

$$\begin{aligned} x + 2x + 6 + 3 &= 60 \\ (1 + 2)x + 9 &= 60 \\ 3x + 9 &= 60 \\ 3x + 9 - 9 &= 60 - 9 \quad \text{[Subtract 9 on both sides]} \\ 3x &= 51 \\ \frac{3x}{3} &= \frac{51}{3} \quad \text{[Divide by 3 on both sides]} \\ \text{Therefore, } x &= 17. \end{aligned}$$

Thus, the number of students got down in the first bus stop is 17.

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